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Introduction to Week 4

Lesson 1: Introduction to Classification

Lesson 2: Logistic Regression

Lesson 3: Logistic Regression, Continued

Video: hypothesis testing in Logistic Regression8 min

Video: Logistic Regression with More Classes2 min

Graded Assignment: Learning Check - Logistic Regression with More ClassesSubmitted

Weekly Assignment

Your grade: 100%

Your latest: 100%

Your highest: 100%

Next item →

Instructions

Review Learning Objectives

1. Suppose we have a dataset of 200 table tennis players at a tournament for which we have access to two features: X_1 = "age", X_2 = "player rating". We define a tournament win as 1 and a loss as 0. We fit a logistic regression and produce the following estimated coefficients: $\beta_0 = -1$, $\beta_1 = 0.01$, and $\beta_2 = 0.0004$. Estimate the probability that a player with age 35 and a rating of 1500 will win the tournament.

07%02%53%89%

Help me practice

Let's chat

Assignment details

DueOct 4, 11:59 PM IST3 attempts every 24 hours

Attempts

Try again

$$p(X) = \frac{e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2}}{1 + e^{\beta_0 + \beta_1 X_1 + \beta_2 X_2}} = \frac{e^{-1 + 0.01(35) + 0.0004(1500)}}{1 + e^{-1 + 0.01(35) + 0.0004(1500)}} = 53\%$$

Your grade

To pass you need at least 80%. We keep your highest score.

2. What would have to be the rating of a player with age 35 to have a 50% chance of winning the tournament? (The answer should be a whole number with four digits and no decimals.)

1625

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Correct

$$\ln\left(\frac{p(X)}{1 - p(X)}\right) = \ln\left(\frac{0.5}{0.5}\right) = 0 = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$
$$= -1 + 0.01(35) + 0.0004(X_2)$$
$$\rightarrow 0.0004(X_2) = 1 - 0.01(35) = 1625$$

3. For the following question, use the table below, which represents the coefficients of a logistic regression for the Default dataset (monetary units in this dataset are US dollars):

	Coefficient	Std. Dev	Z-statistic
Intercept	-10.8690	0.4923	-22.08
Balance	0.0057	0.0002	24.74
Income	0.0030	0.0082	0.37
Student (yes)	-0.6468	0.2362	-2.74

What conclusions can you draw based on the table? Use a 95% confidence level to answer this question. (Select all that apply.)

Note: The book and standard libraries for logistic regression use the Z-statistic. It is calculated by dividing the coefficient by the standard error (or standard deviation), just like the t-statistic. In essence, the Z-statistic is the t-statistic for logistic regression.

t-statistic: Linear Regression Z-statistic: Logistic Regression

However, note that the Z-statistic is **not** the z-score, which is used in statistics to denote the right tail quantile of a normal distribution.

You can read more [here](#).

☐ Whether a borrower is a student is not statistically significant to whether one defaults.

☒ For every \$333.333 dollar increase in income, the log-odds function increases by 1.

Correct

This interpretation of logistic regression is that for every dollar increase in income, the log-odds function increases by 0.0030. Therefore, if there is a \$333.33 dollar increase in income, the log-odds function increases by $333.33 \times 0.0030 = 1333.35$ times $0.0030 = 1333.33 \times 0.0030 = 1$

☐ For every half-cent increase in balance, the probability of default increases by 1%.

☐ The null hypothesis is that balance influences the output.

☒ How much income one has is not statistically significant to whether one defaults.

Correct

The magnitude of the Z-statistic is less than 2, so it is not significant.

https://www.coursera.org/learn/machine-learning-essentials/assignment-submission/Eoy4K/learning-check-logistic-regression-with-more-classes/view-feedback

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